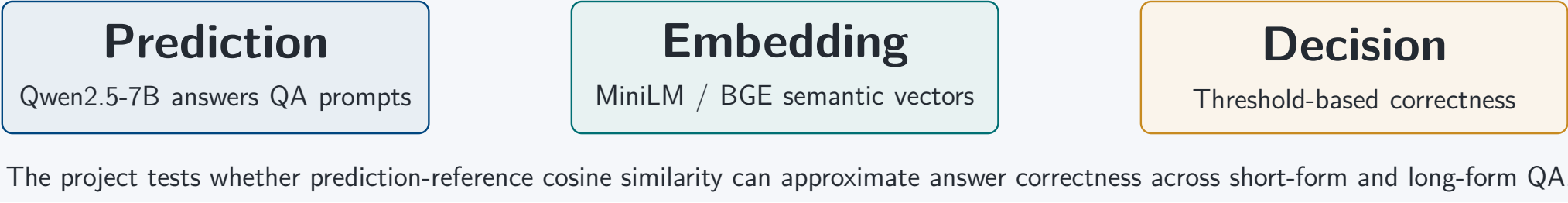


Semantic Similarity in Latent Space for LLM Prediction Evaluation

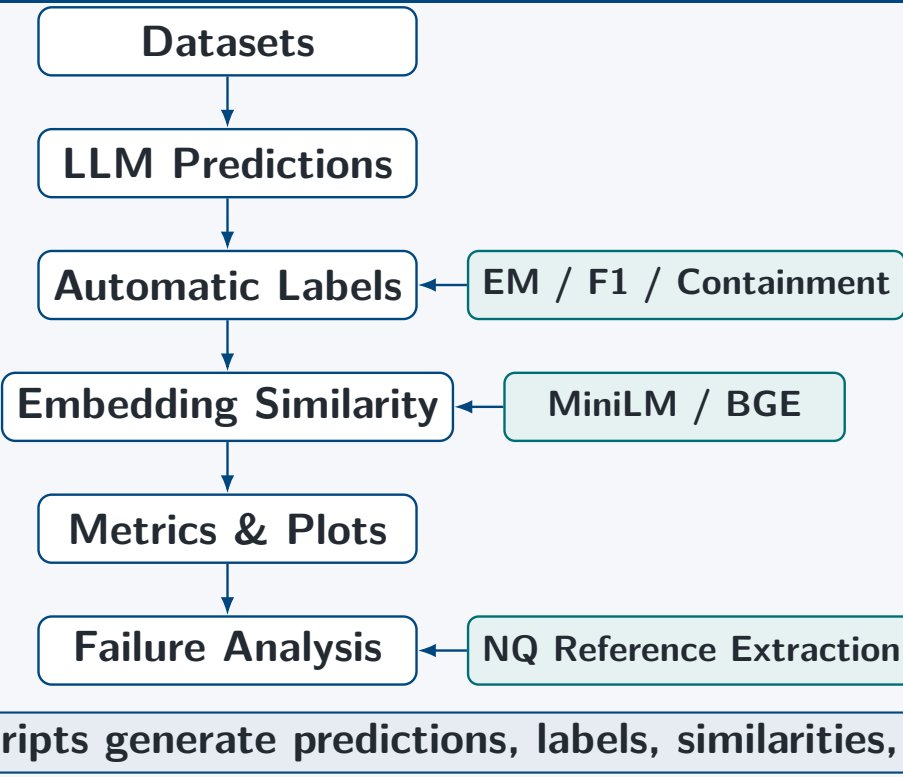
Team Members: Alice, Bob, Charlie, David | AIAA 4051 Final Research Project, Introduction to NLP, HKUST(GZ)

Project Goal

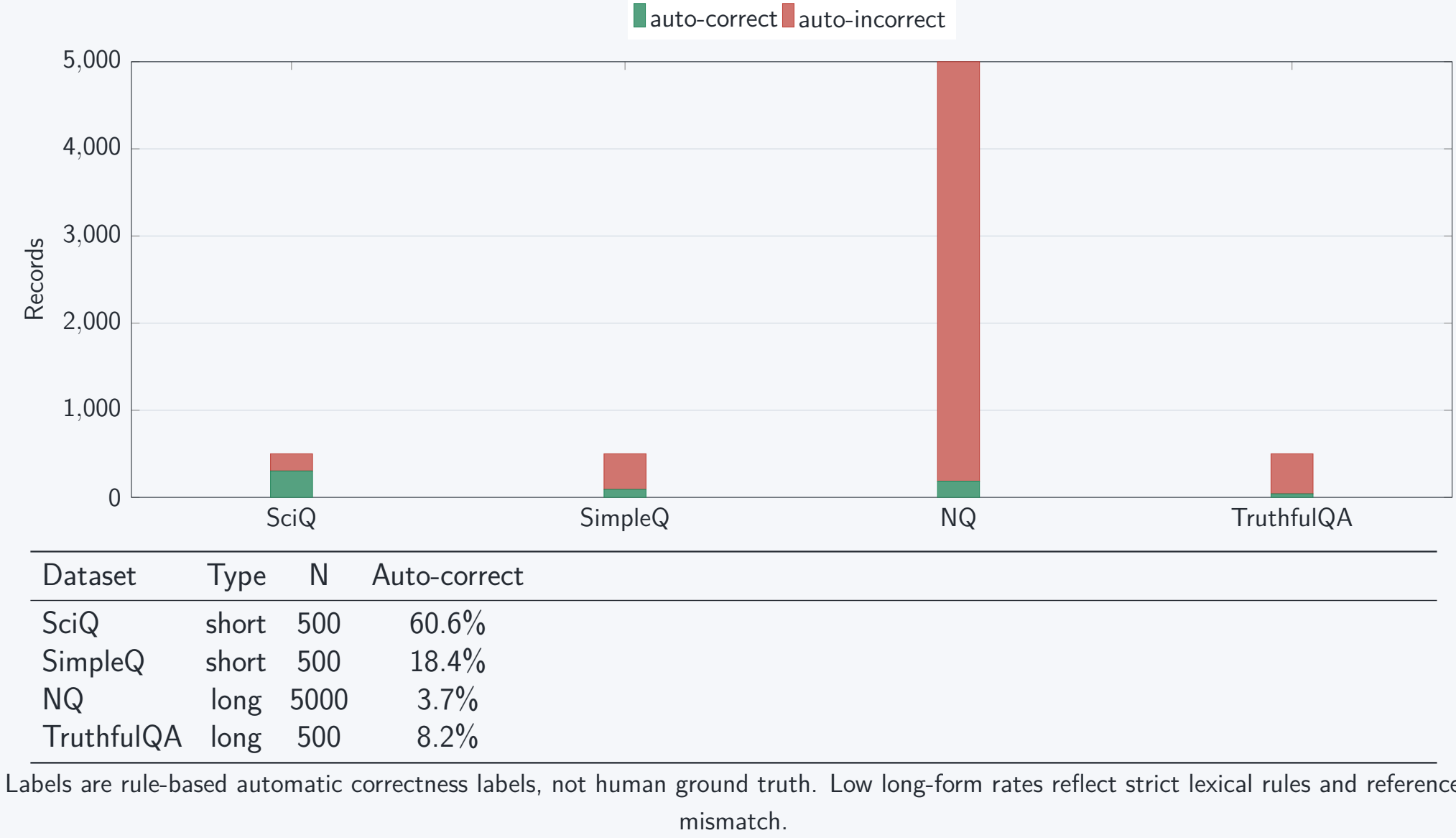
Latent-space embedding similarity as an automatic QA correctness signal



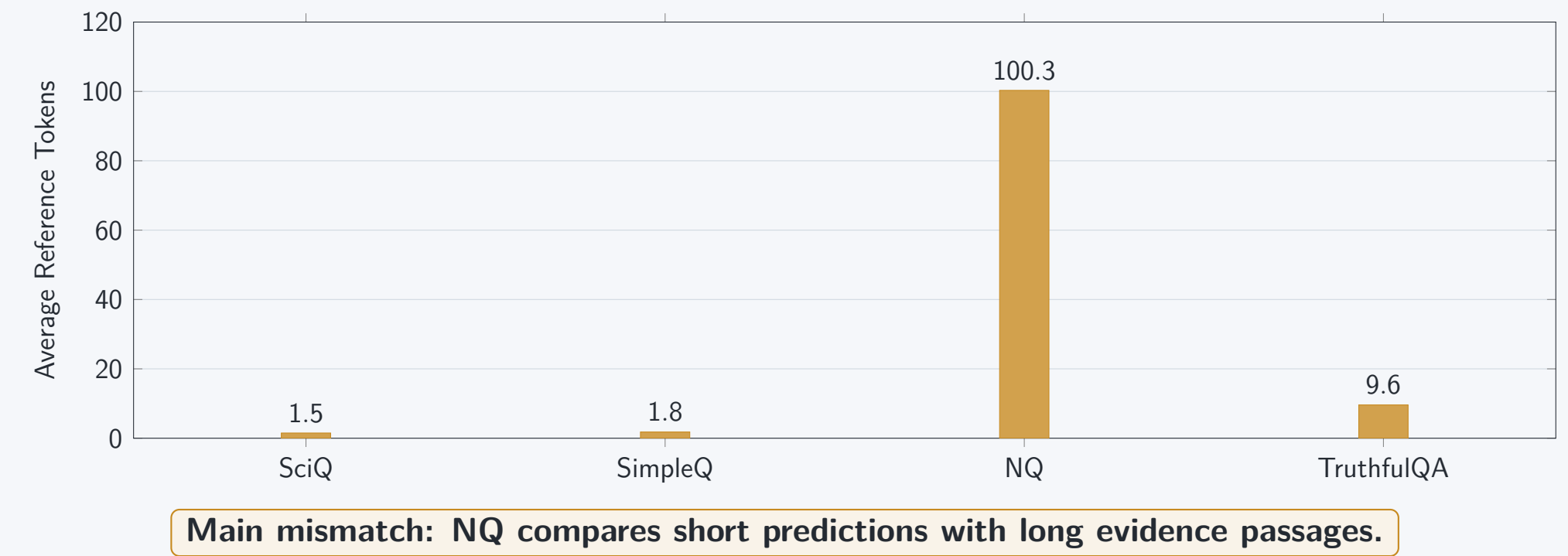
Pipeline Overview



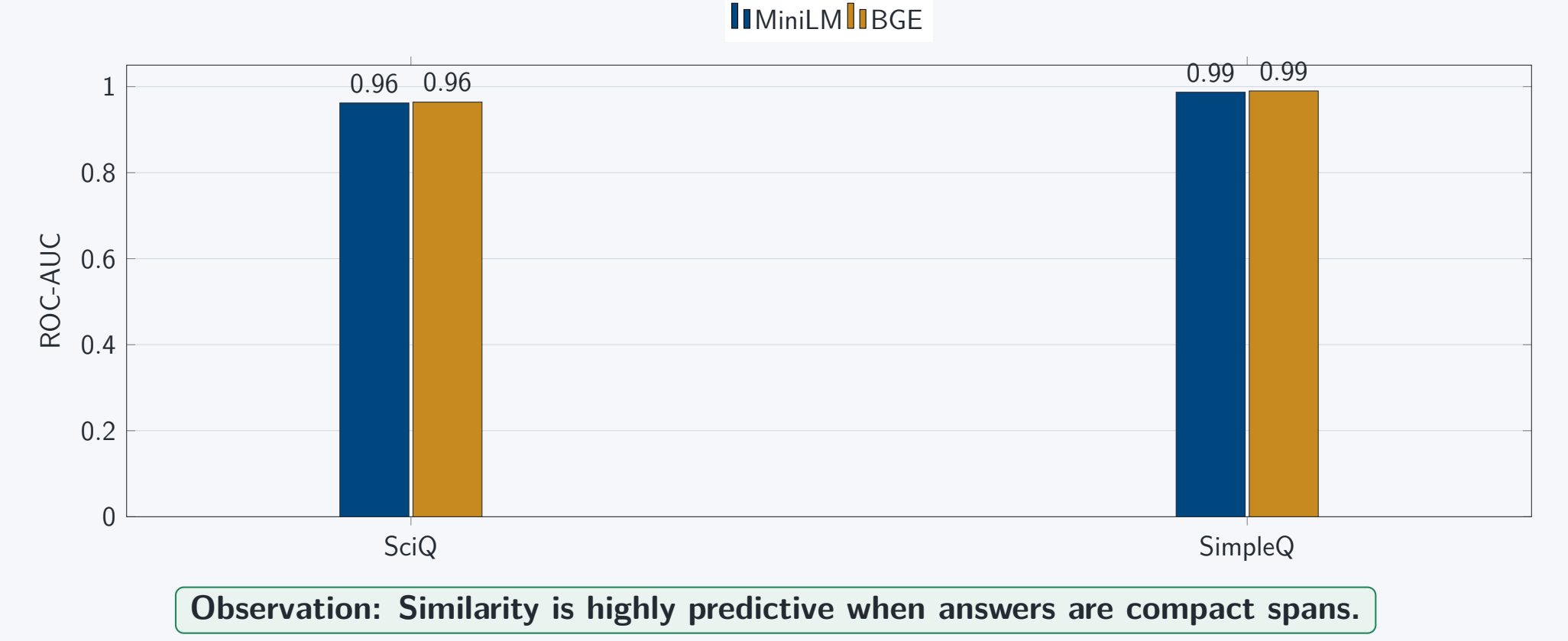
Datasets & Automatic Label Distribution



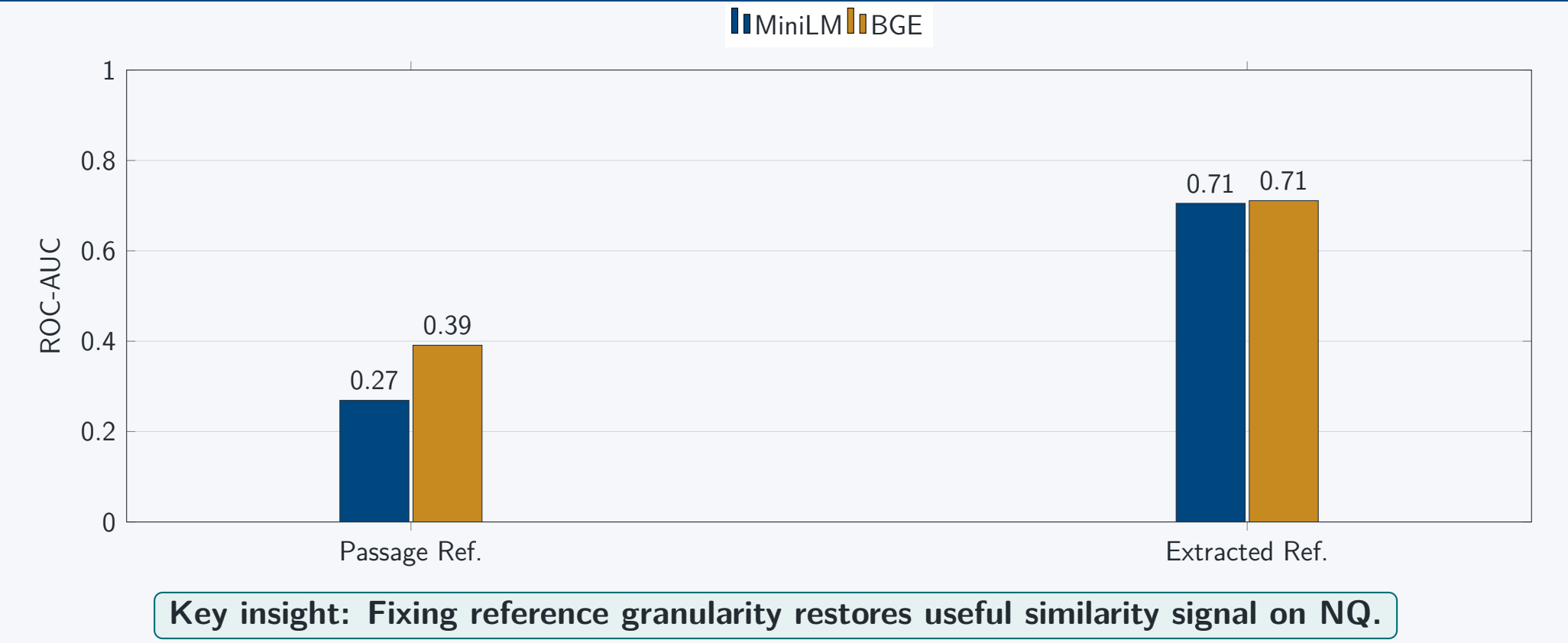
Reference Length Mismatch



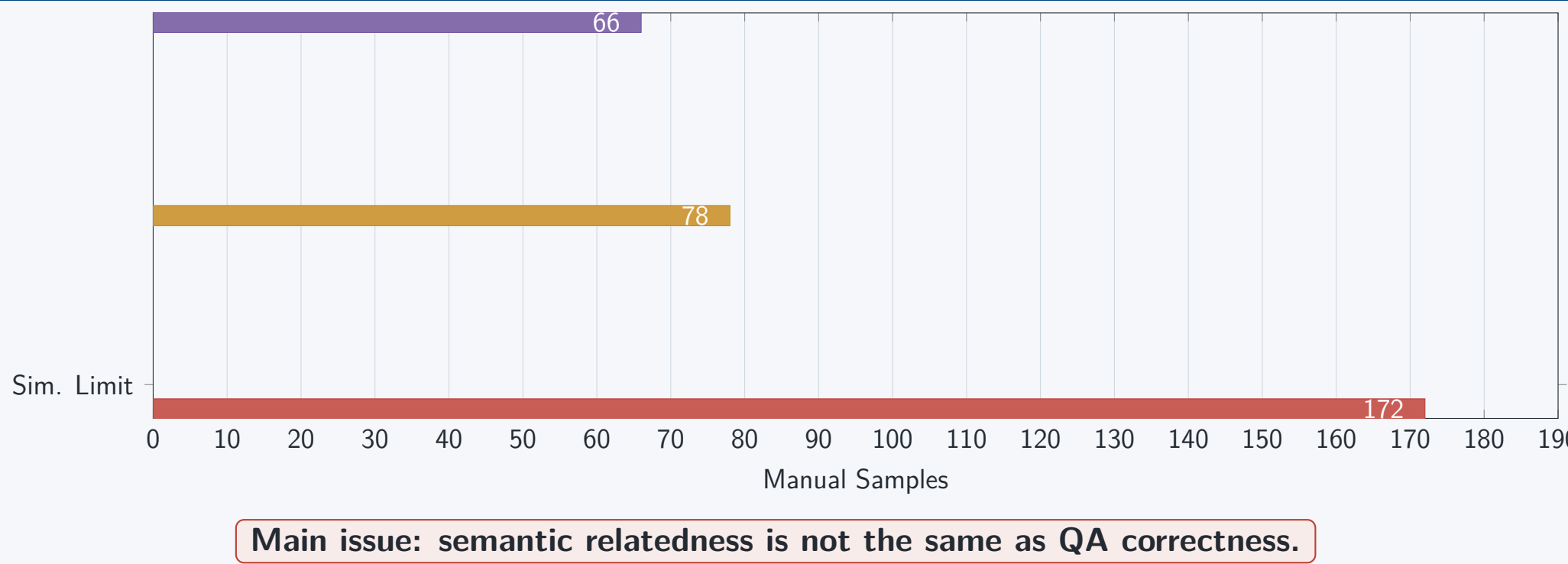
ROC-AUC: Short QA



ROC-AUC: NQ Reference Extraction



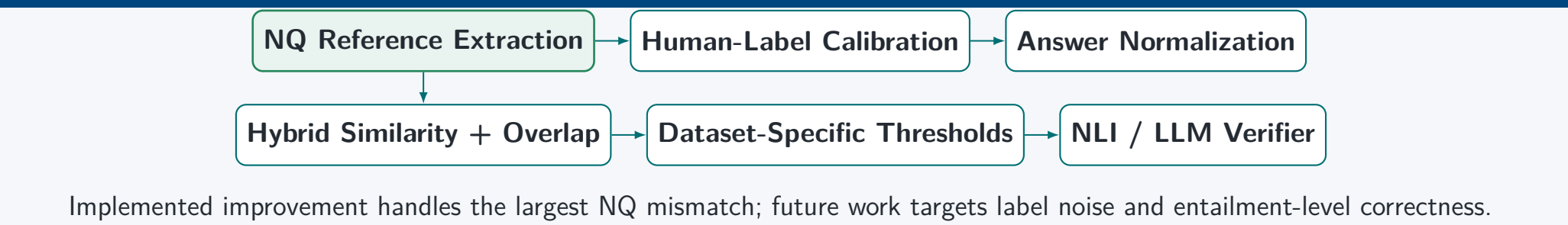
Failure Analysis: Manual Categories



Failure Analysis: What the Errors Mean

Topic overlap same passage topic	Granularity too broad / specific	Label artifact alias or paraphrase	Dilution answer hidden in text
High-sim wrong	Often caused by topical relatedness, long passages, or semantically close but incorrect answers.		
Low-sim correct	Usually caused by short-answer vs. long-context mismatch or extra generated context.		
Label artifact	Automatic rules are stricter than human judgment for aliases and paraphrases.		

Improvement Roadmap



Final Conclusion

Works short answer units	Breaks long reference mismatch	Improves extract + verify
Embedding similarity is a useful proxy only when prediction and reference are comparable answer units.		